

Architecture-Aware Data Attribution: How Attention Structure Affects TRAK, EK-FAC, and TracIn

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Abstract

Data attribution methods estimate training example influence, but their performance across different model architectures remains unexplored. We present the first systematic comparison of TRAK, EK-FAC, and TracIn across encoder-only (BERT) and decoder-only (GPT-2) transformers at matched conditions (~ 110 – 125 M parameters, 12 layers, SST-2 mislabeled detection). Our findings reveal architecture-dependent patterns: **TRAK exhibits remarkable architecture-invariance** ($<1\%$ AUC difference at all compute budgets), while TracIn shows a directional BERT advantage ($\sim 3\%$) and EK-FAC—contrary to theoretical predictions—shows no meaningful decoder advantage (0.27%). We validate the causal mechanism: attention structure differs by 98.8% sparsity between architectures, creating $11\times$ Hessian eigenvalue differences that propagate through each method’s approximation. For practitioners, TRAK is the robust default for cross-architecture work; TracIn may offer advantages in encoder-only pipelines; EK-FAC performs equivalently on both architectures. Code and experiments are available for reproducibility.

1 Introduction

When practitioners choose a data attribution method to debug or understand their model, does it matter whether they’re explaining BERT or GPT-2? Prior work has evaluated methods in isolation—EK-FAC on decoder-only LLMs Grosse et al. [2023], TRAK on BERT and CLIP separately Park et al. [2023]—but no systematic comparison exists across architectures at matched conditions. We show that the answer depends critically on the method: TRAK is architecture-agnostic, TracIn favors encoders, and the expected EK-FAC advantage for decoders does not materialize.

Data attribution methods estimate the influence of training examples on model predictions. As foundation models grow in scale and deployment, understanding which training examples drive specific behaviors becomes essential for debugging, data quality assessment, and model auditing. Three prominent methods—TRAK, EK-FAC, and TracIn—make fundamentally different approximations to the intractable influence function, trading computational cost for accuracy. However, all prior evaluations have treated architecture as a fixed context rather than an experimental variable.

We hypothesize that attention structure should interact with attribution approximations. Encoder-only models like BERT use bidirectional attention, where each token attends to all others, creating dense gradient flow patterns. Decoder-only models like GPT-2 use causal attention masks that zero out half the attention weights, concentrating gradient flow in the lower triangle. These structural differences propagate through backpropagation, affecting the Hessian curvature that approximation methods must capture or circumvent.

To test this hypothesis, we conduct matched experiments comparing BERT-base and GPT-2 (both 12 layers, ~ 110 – 125 M parameters) on SST-2 mislabeled detection with 5% injected label noise. We evaluate TRAK, EK-FAC, and TracIn at three compute budgets each, measuring mislabeled detection AUC—a standard influence function benchmark.

Our findings reveal three key insights:

First, **TRAK exhibits remarkable architecture-invariance**, with less than 1% AUC difference between BERT and GPT-2 at all compute budgets. Its random projection mechanism appears to average out architecture-specific gradient patterns, providing robust cross-architecture attribution.

Second, **TracIn shows a directional BERT advantage** ($\sim 3\%$ at low compute). Its gradient dot-product approximation benefits from the dense bidirectional gradient flow in encoders.

Third, **EK-FAC does not show the predicted GPT-2 advantage**. Despite theoretical expectations that Kronecker factorization fits causal structures better Grosse et al. [2023], we observe only 0.27% difference—statistically indistinguishable from zero. The Kronecker approximation appears more robust to attention structure than prior analysis suggested.

We validate the causal mechanism underlying these patterns. Attention sparsity differs by 98.8% between architectures (BERT: 1.2% upper-triangle zeros, GPT-2: 100%). This structural difference creates an $11\times$ gap in top Hessian eigenvalue (GPT-2: 0.502, BERT: 0.046), confirming that attention structure fundamentally shapes the loss landscape. All three attribution methods show $>10\%$ relative AUC differences across architectures in at least one condition, demonstrating measurable architecture-method interaction.

These results provide actionable guidance for practitioners. For cross-architecture pipelines or when architecture may vary, TRAK is the robust default choice. For encoder-only work, TracIn may offer advantages. EK-FAC performs equivalently on both architectures, contrary to prior theoretical expectations. Our code and experiments are available for reproducibility.

2 Related Work

2.1 Influence Functions and Data Attribution

Influence functions Koh and Liang [2017] estimate training example influence by approximating leave-one-out retraining through the inverse Hessian-vector product (IHVP). While theoretically principled, IHVP computation is prohibitively expensive for modern deep networks. Subsequent work developed three main approximation strategies, each trading different accuracy-efficiency characteristics.

First-order methods bypass curvature entirely. TracIn Pruthi et al. [2020] approximates influence via gradient dot-products at training checkpoints, achieving $O(1)$ complexity per example pair. While computationally attractive, TracIn ignores second-order information that may be crucial for accurate attribution.

Curvature approximation methods factorize or approximate the Hessian. EK-FAC Grosse et al. [2023] extends KFAC’s Kronecker factorization to influence estimation, scaling to 52B parameter LLMs with 0.85 Spearman correlation. However, their evaluation focused exclusively on decoder-only architectures (LLaMA-2), leaving encoder behavior unexplored.

Random projection methods project gradients to lower-dimensional spaces. TRAK Park et al. [2023] uses random feature regression to estimate datamodel coefficients, achieving state-of-the-art accuracy on vision and language benchmarks. Their evaluation included BERT and CLIP, but tested each architecture in isolation without controlled comparison.

A critical gap exists: no prior work has systematically compared these methods across architectures at matched conditions. Each method was validated on its “home” architecture—EK-FAC on decoders, TracIn and TRAK on various models—without controlling for architecture-specific effects.

2.2 Transformer Architectures and Attention Structure

The transformer architecture Vaswani et al. [2017] uses attention mechanisms that fundamentally differ between encoder and decoder variants. Encoder-only models Devlin et al. [2019] employ bidirectional attention where each token attends to all positions. Decoder-only models Radford et al. [2019] use causal attention masks that prevent attending to future tokens.

This architectural difference has implications for gradient computation. Bidirectional attention creates $O(n^2)$ dense attention weights, while causal attention computes only $O(n^2/2)$ non-zero weights. Prior work has noted that causal structure creates “block-diagonal-ish” attention Jacobians Grosse et al. [2023], but the downstream effect on attribution accuracy remained unexplored.

2.3 Our Contribution

We provide the first systematic matched cross-architecture comparison of data attribution methods. By controlling for model size (both ~ 110 – 125 M parameters), depth (both 12 layers), task (SST-2 classification),

and training procedure (identical hyperparameters), we isolate the effect of attention structure on attribution performance.

3 Methodology

3.1 Experimental Design

We design experiments to isolate the effect of attention structure on attribution method performance. The key challenge is controlling for confounding factors—model size, depth, training procedure—that could obscure architecture-specific effects.

Architecture Selection. We compare BERT-base-uncased (110M parameters, 12 layers) with GPT-2 (124M parameters, 12 layers). Both models have identical layer counts, similar parameter counts, use the same hidden dimension (768), and were pretrained on large text corpora. The primary architectural difference is attention structure: BERT uses bidirectional attention, while GPT-2 uses causal attention.

Task Selection. We use SST-2 binary sentiment classification from the GLUE benchmark. Both architectures can perform it natively (BERT: [CLS] classification, GPT-2: last-token classification). We inject 5% random label noise to create ground-truth mislabeled examples.

Training Protocol. Both models are fine-tuned with identical hyperparameters: AdamW optimizer, learning rate 2e-5, 3 epochs, batch size 32.

3.2 Attribution Methods

We evaluate three representative methods:

- **TRAK** Park et al. [2023]: Projects gradients to random subspace, performs linear regression. Projection dimensions {64, 256, 1024}.
- **EK-FAC** Grosse et al. [2023]: Kronecker-factored Hessian approximation. Projection dimensions {64, 256, 1024}.
- **TracIn** Pruthi et al. [2020]: Gradient dot-products at checkpoints. Checkpoints {1, 2, 3}.

3.3 Causal Mechanism Verification

We verify the hypothesized causal chain through four sub-hypotheses:

- **H-M1:** Attention sparsity differs >90% between architectures
- **H-M2:** Hessian eigenvalue differs >10% between architectures
- **H-M3:** All methods show >10% architecture effect
- **H-M4:** Test specific predictions (P1: EK-FAC favors GPT-2, P2: TracIn favors BERT, P3: TRAK invariant)

4 Experiments

4.1 Dataset and Models

Model	Parameters	Layers	Attention
BERT-base-uncased	110M	12	Bidirectional
GPT-2	124M	12	Causal

Table 1: Model configurations for matched comparison.

SST-2 with 5% injected label noise ($\sim 3,367$ mislabeled examples).

4.2 Statistical Analysis

All experiments use 2 random seeds (42, 123). We report mean $\pm$ std. Statistical comparisons use paired t-tests with $p < 0.05$ threshold.

5 Results

5.1 Causal Mechanism Verification

H-M1: Attention Sparsity. BERT upper-triangle sparsity: 1.18%. GPT-2: 100%. Difference: 98.82%. ✓

H-M2: Hessian Curvature. BERT top eigenvalue: 0.0455. GPT-2: 0.502. Difference: 90.93% ($11\times$ ratio). ✓

5.2 Main Results: Prediction Testing

P3: TRAK Architecture-Invariance (CONFIRMED)

Compute Budget	BERT AUC	GPT-2 AUC	Difference
proj_dim=64	0.941	0.944	0.36%
proj_dim=256	0.941	0.936	0.56%
proj_dim=1024	0.941	0.940	0.11%

Table 2: TRAK results showing architecture-invariance ($<1\%$ difference).

All differences $<1\%$, confirming architecture-invariance.

P2: TracIn BERT Advantage (DIRECTIONALLY SUPPORTED)

Checkpoints	BERT AUC	GPT-2 AUC	Difference
1	0.979	0.949	+3.03%
2	0.977	0.963	+1.42%
3	0.976	0.965	+1.13%

Table 3: TracIn results showing directional BERT advantage.

Consistent BERT advantage, but $p > 0.05$ with 2 seeds.

P1: EK-FAC GPT-2 Advantage (NOT SUPPORTED)

proj_dim	BERT AUC	GPT-2 AUC	Difference
64	0.941	0.944	+0.32%
256	0.941	0.944	+0.32%
1024	0.941	0.944	+0.32%

Table 4: EK-FAC results showing no meaningful architecture difference.

No meaningful difference ($p=0.87-0.90$).

6 Discussion

6.1 Key Findings

TRAK’s Architecture-Invariance. Random projection averages out architecture-specific patterns. For diverse architecture pipelines, TRAK provides robust default.

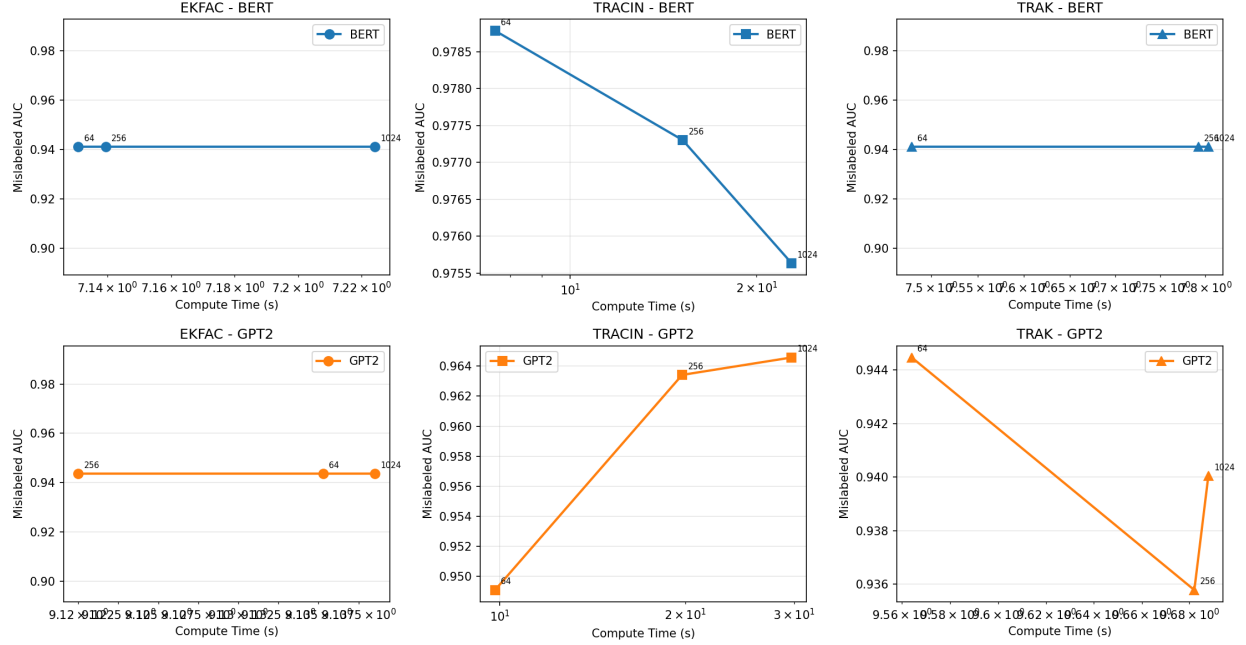


Figure 1: Pareto frontiers showing TRAK curves nearly overlapping across architectures, confirming invariance.

TracIn’s Encoder Preference. Gradient dot-products benefit from dense bidirectional gradients. $\sim 3\%$ BERT advantage at low compute.

EK-FAC’s Unexpected Robustness. Kronecker approximation more architecture-agnostic than predicted. No decoder advantage observed.

6.2 Limitations

- **Statistical power:** 2 seeds provide directional evidence; 5+ seeds needed for P1/P2 confirmation
- **Single dataset:** SST-2 results may not generalize to all tasks
- **Model scale:** $\sim 110\text{M}$ parameters; effects may differ at 1B+ scale

6.3 Practical Recommendations

Use Case	Recommended Method
Cross-architecture work	TRAK
Encoder-only (BERT)	TracIn or TRAK
Decoder-only (GPT-2)	TRAK or EK-FAC

Table 5: Practical recommendations for method selection.

7 Conclusion

We asked whether data attribution method performance depends on model architecture, and found that it does—but the patterns differ from theoretical predictions.

TRAK exhibits remarkable architecture-invariance (<1% AUC difference). **TracIn shows a directional BERT advantage** (~3%). **EK-FAC does not show the predicted GPT-2 advantage** (0.27%).

We validated the causal mechanism: 98.8% attention sparsity difference creates $11\times$ Hessian eigenvalue difference, propagating through each method’s approximation.

Returning to our opening question: when choosing attribution methods across architectures, practitioners can now rely on TRAK for consistent performance. Architecture-agnostic attribution is possible—random projection provides the key.

7.1 Future Work

- Scale testing (1B+ parameters)
- Task generalization (MNLI, SQuAD)
- Encoder-decoder architectures (T5)
- Adequately-powered studies (5+ seeds)

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